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Roll No. .....

IIIrd Sem.

B.Tech. (PT)

**SUPPLEMENTARY EXAMINATION**

FEB-2018

**CEEE-301 ENGINEERING ECONOMICS**

Time: 3:00 Hours

Max. Marks : 70

**Note :** Answer any *FIVE* questions.

Assume suitable missing data, if any.

All questions carry equal marks.

1[a] What is meant by demand? Also discuss the various determinants of consumers demand. 7

[b] Define price elasticity of demand and discuss various methods of measuring price elasticity of demand. 7

2 [a] Explain the law of diminishing marginal utility. Illustrate your answer with appropriate diagrams. Discuss also the assumptions and limitations of this law. 7

[b] A company earned a profit of Rs.60000 during the current year. If the average variable cost and selling price of a product is Rs.20 and Rs.25 per unit respectively, find out the margin of safety. 7

3[a] Given the demand function and total cost function of a perfectly competitive firm as: 7

$$P = 120 - X$$

$$TC = X^3 - X^2 - 10X + 2$$

What level of output will maximize total profit and what are the corresponding values of price (P), profit and Total Revenue?

[b] What do you mean by sales forecasting? Discuss various methods of sales forecasting. 7

4[a] Ms Kavita deposits her annual bonus into a saving account that pays 8% interest compounded annually. The size of her bonus increases by Rs.5000 each year, and the

initial bonus amount is Rs.500. Determine how much will be in the account immediately after the seventh deposit. 7

[b] The cash flows of two projects proposals are as given below. Each of the projects has an expected life of 10 years. Select the best project based on present Worth method of comparison using an interest rate of 18% compounded annually. 7

| | Initial outlay(Rs.) | Annual equivalent revenue(Rs) | Salvage value after 10 years(Rs.) |
|---|---|---|---|
| Project 1 | - 7,50,000 | 2,00,000 | 50,000 |
| Project 2 | -9,50,000 | 2,25,000 | 1,00,000S |

5[a] A company wants to replace a present facility after 15 years at an outlay of Rs. 5,00,000. It plans to deposit an equal amount at the end of every year for the next 15 years at an interest rate of 18% compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 15 years. 7

[b] A person who is 30 years old is planning for retired life. He plans to invest an equal sum of Rs. 15,000 at the end of every year for next 25 years starting from the end of the next year. The bank gives 12% interest rate, compounded annually. Find the maturity value of his account when he is 55 years old. 7

6[a] A Bank gives a loan to a company to purchase an equipment worth Rs. 10,00,000 at an interest rate of 18% compounded annually. This amount should be repaid in 15 yearly equal installments. Find the installment amount that the company has to pay to the bank. 7

[b] A company must decide whether to buy machine A or machine B: 7

| | Machine A | Machine B |
|---|---|---|
| Initial cost (Rs.) | 4,00,000 | 8,00,000 |
| Useful life (years) | 5 | 5 |
| Salvage value at the end of machine life (Rs.) | 2,00,000 | 5,00,000 |
| Annual maintenance cost(Rs) | 40,000 | 0 |

At 15% interest rate which machine should be selected?.Use the future Worth method c comparison

END